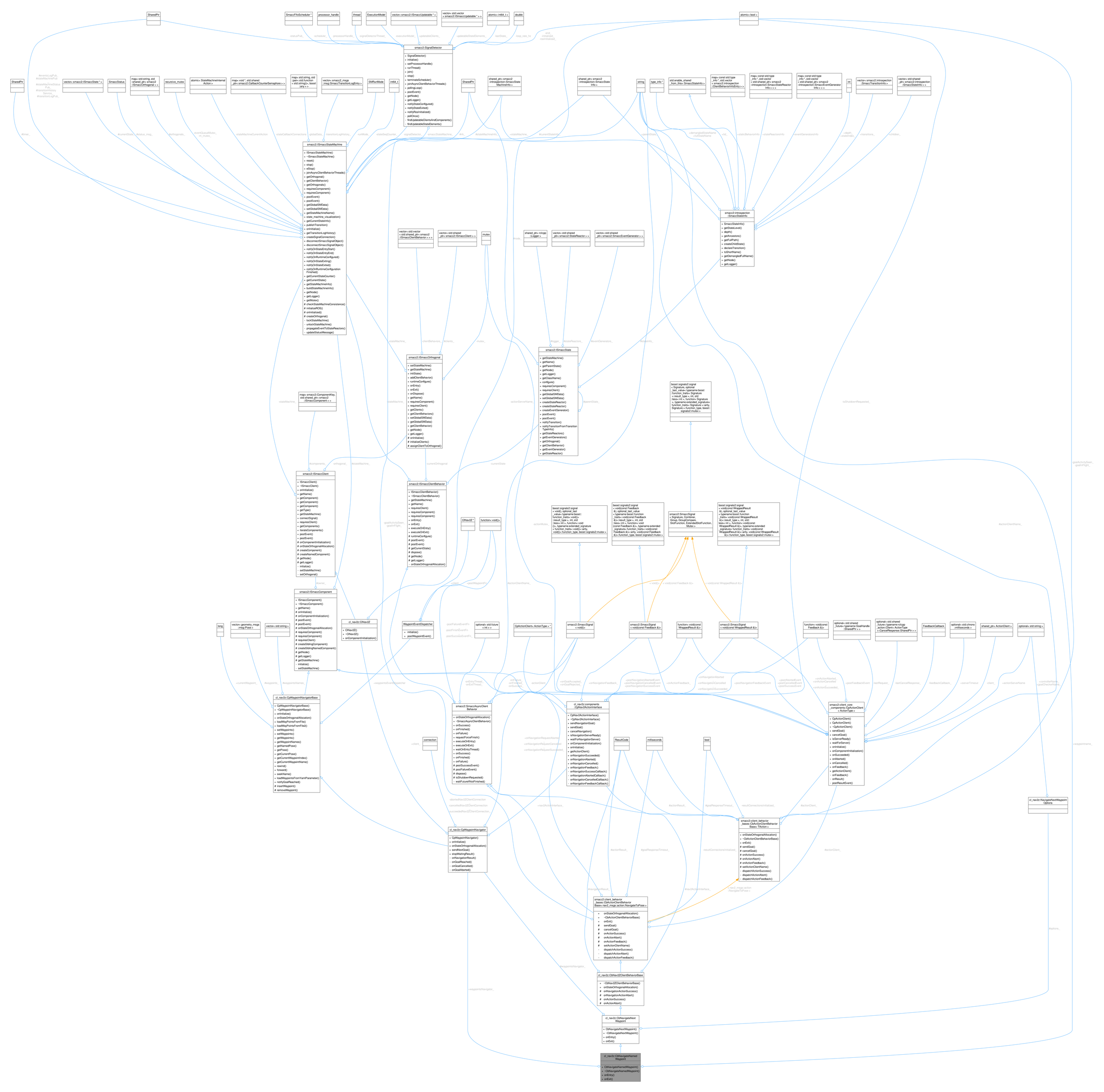
SharedPtr
SmaccFifoScheduler *
processor_handle
thread
ExecutionModel
vector< smacc2::ISmaccUpdatable * >
vector< std::vector < smacc2::ISmaccUpdatable * > >
atomic< int64_t >
double
atomic< bool >
smacc2::SignalDetector
+ SignalDetector()
+ initialize()
+ setProcessorHandle()
+ runThread()
+ join()
+ stop()
+ terminateScheduler()
+ joinAsyncClientBehaviorThreads()
+ pollingLoop()
+ postEvent()
+ getNode()
+ getLogger()
+ notifyStateConfigured()
+ notifyStateExited()
+ notifyRosInitialized()
- pollOnce()
- findUpdatableClientsAndComponents()
- findUpdatableStateElements()
SharedPtr
vector< smacc2::ISmaccState * >
SmaccStatus
map< std::string, std ::shared_ptr< smacc2 ::ISmaccOrthogonal > >
recursive_mutex
atomic< StateMachineInternal Action >
map< void *, std::shared _ptr< smacc2::CallBackCounterSemaphore > >
map< std::string, std ::pair< std::function < std::string() >, boost ::any > >
vector< smacc2_msgs ::msg::SmaccTransitionLogEntry >
SMRunMode
int64_t
SharedPtr
shared_ptr< smacc2 ::introspection::SmaccState MachineInfo >
shared_ptr< smacc2 ::introspection::SmaccState Info >
string
type_info *
std::enable_shared _from_this< SmaccStateInfo >
map< const std::type _info *, std::vector < smacc2::introspection ::ClientBehaviorInfoEntry > >
map< const std::type _info *, std::vector < std::shared_ptr< smacc2 ::introspection::SmaccStateReactor Info > > >
map< const std::type _info *, std::vector < std::shared_ptr< smacc2 ::introspection::SmaccEventGenerator Info > > >
int
vector< smacc2::introspection ::SmaccTransitionInfo >
vector< std::shared _ptr< smacc2::introspection ::SmaccStateInfo > >
#timer_
#currentState_
#status_msg_
#orthogonals_
#eventQueueMutex_
#n_mutex_
#stateMachineCurrentAction
#stateCallbackConnections_
#globalData_
#transitionLogHistory_
#runMode_
#stateSeqCounter_
#signalDetector_
#smaccStateMachine_
#this_
#stateMachineInfo_
#stateMachine_
#currentStateInfo_
#parentState_
#demangledStateName
#fullStateName
#id_
#staticBehaviorInfo_
#stateReactorsInfo
#eventGeneratorsInfo
#depth_
#stateIndex_
#transitions_
#children_
smacc2::ISmaccStateMachine
+ ISmaccStateMachine()
+ ~ISmaccStateMachine()
+ reset()
+ stop()
+ eStop()
+ joinAsyncClientBehaviorThreads()
+ getOrthogonal()
+ getClientBehavior()
+ getOrthogonals()
+ requiresComponent()
+ requiresComponent()
+ postEvent()
+ postEvent()
+ getGlobalSMData()
+ setGlobalSMData()
+ getStateMachineName()
+ state_machine_visualization()
+ getCurrentStateInfo()
+ publishTransition()
+ onInitialize()
+ getTransitionLogHistory()
+ createSignalConnection()
+ disconnectSmaccSignalObject()
+ disconnectSmaccSignalObject()
+ notifyOnStateEntryStart()
+ notifyOnStateEntryEnd()
+ notifyOnRuntimeConfigured()
+ notifyOnStateExiting()
+ notifyOnStateExited()
+ notifyOnRuntimeConfiguration Finished()
+ getCurrentStateCounter()
+ getCurrentState()
+ getStateMachineInfo()
+ buildStateMachineInfo()
+ getNode()
+ getLogger()
+ getMutex()
# checkStateMachineConsistence()
# initializeROS()
# onInitialized()
# createOrthogonal()
- lockStateMachine()
- unlockStateMachine()
- propagateEventToStateReactors()
- updateStatusMessage()
smacc2::introspection ::SmaccStateInfo
+ SmaccStateInfo()
+ getStateLevel()
+ depth()
+ getAncestors()
+ getFullPath()
+ createChildState()
+ declareTransition()
+ toShortName()
+ getDemangledFullName()
+ getNode()
+ getLogger()
vector< std::vector < std::shared_ptr< smacc2 ::ISmaccClientBehavior > > >
vector< std::shared _ptr< smacc2::ISmaccClient > >
mutex
shared_ptr< rclcpp ::Logger >
vector< std::shared _ptr< smacc2::StateReactor > >
vector< std::shared _ptr< smacc2::SmaccEventGenerator > >
#mode_
#logger_
#stateReactors_
#eventGenerators_
#stateInfo_
#stateMachine_
#clientBehaviors_
#clients_
#mutex_
smacc2::ISmaccOrthogonal
+ setStateMachine()
+ getStateMachine()
+ initState()
+ addClientBehavior()
+ runtimeConfigure()
+ onEntry()
+ onExit()
+ onDispose()
+ getName()
+ requiresComponent()
+ requiresClient()
+ getClients()
+ getClientBehaviors()
+ setGlobalSMData()
+ getGlobalSMData()
+ getClientBehavior()
+ getNode()
+ getLogger()
# onInitialize()
# initializeClients()
# assignClientToOrthogonal()
smacc2::ISmaccState
+ getStateMachine()
+ getName()
+ getParentState()
+ getNode()
+ getLogger()
+ getClassName()
+ configure()
+ requiresComponent()
+ requiresClient()
+ getGlobalSMData()
+ setGlobalSMData()
+ createStateReactor()
+ createStateReactor()
+ createEventGenerator()
+ postEvent()
+ postEvent()
+ notifyTransition()
+ notifyTransitionFromTransition TypeInfo()
+ getStateReactors()
+ getEventGenerators()
+ getOrthogonal()
+ getClientBehavior()
+ getEventGenerator()
+ getStateReactor()
boost::signals2::signal
< Signature, optional
_last_value= typename boost
::function_traits< Signature
>::result_type >, int, std
::less< int >, function< Signature
>, typename extended_signature<
function_traits< Signature >::arity,
Signature >::function_type, boost::
signals2::mutex >
map< smacc2::ComponentKey, std::shared_ptr< smacc2 ::ISmaccComponent > >
#components_
#orthogonal_
#stateMachine_
#currentOrthogonal_
#currentState_
#actionServerName_
#eventState_
#isShutdownRequested_
#goalActivityStateen_
#goalInFlight_
smacc2::ISmaccClient
+ ISmaccClient()
+ ~ISmaccClient()
+ onInitialize()
+ getName()
+ getComponent()
+ getComponent()
+ getComponent()
+ getType()
+ getStateMachine()
+ connectSignal()
+ requiresClient()
+ getComponents()
+ iterateComponents()
+ postEvent()
+ postEvent()
# onComponentInitialization()
# onStateOrthogonalAllocation()
# createComponent()
# createNamedComponent()
# getNode()
# getLogger()
- initialize()
- setStateMachine()
- setOrthogonal()
smacc2::ISmaccClientBehavior
+ ISmaccClientBehavior()
+ ~ISmaccClientBehavior()
+ getStateMachine()
+ getName()
+ requiresClient()
+ requiresComponent()
+ requiresComponent()
+ onEntry()
+ onExit()
+ executeOnEntry()
+ executeOnExit()
# runtimeConfigure()
# postEvent()
# postEvent()
# getCurrentState()
# dispose()
# getNode()
# getLogger()
- onStateOrthogonalAllocation()
boost::signals2::signal
< void(), optional_last
_value< typename boost::
function_traits< void()>
::result_type >, int, std
::less< int >, function< void
()>, typename extended_signature
< function_traits< void()>::arity,
void()>::function_type, boost::signals2::mutex >
boost::signals2::signal
< void(const Feedback
&), optional_last_value
< typename boost::function
_traits< void(const Feedback
&)>::result_type >, int, std
::less< int >, function< void
(const Feedback &)>, typename extended
_signature< function_traits< void(const
Feedback &)>::arity, void(const Feedback
&)>::function_type, boost::signals2::mutex >
smacc2::SmaccSignal
< Signature, Combiner,
Group, GroupCompare,
SlotFunction, ExtendedSlotFunction,
Mutex >
boost::signals2::signal
< void(const WrappedResult
&), optional_last_value
< typename boost::function
_traits< void(const WrappedResult
&)>::result_type >, int, std::
less< int >, function< void(const
WrappedResult &)>, typename extended
_signature< function_traits< void(const
WrappedResult &)>::arity, void(const WrappedResult
&)>::function_type, boost::signals2::mutex >
#actionClientName_
QNav2Z *
function< void()>
#actionMutex_
< void()>
< void(const Feedback &)>
< void(const WrappedResult &)>
smacc2::ISmaccComponent
+ ISmaccComponent()
+ ~ISmaccComponent()
+ getName()
# onInitialize()
# onComponentInitialization()
# postEntry()
# postEvent()
# onStateOrthogonalAllocation()
# requiresComponent()
# requiresComponent()
# requiresClient()
# createSiblingComponent()
# createSiblingNamedComponent()
# getNode()
# getLogger()
# getStateMachine()
- initialize()
- setStateMachine()
cl_nav2z::CNav2Z
+ CNav2Z()
+ ~CNav2Z()
+ onComponentInitialization()
WaypointEventDispatcher
+ initialize()
+ postWaypointEvent()
#owner_
#client_
#postWaypointFn_
#postFailureEventFn
#postFinishEventFn
#postSuccessEventFn
optional< std::future < int > >
CpActionClient< ActionType > *
smacc2::SmaccSignal
< void()>
smacc2::SmaccSignal
< void(const Feedback &)>
function< void(const WrappedResult &)>
smacc2::SmaccSignal
< void(const WrappedResult &)>
function< void(const Feedback &)>
optional< std::shared _future< typename GoalHandle ::SharedPtr > >
optional< std::shared _future< typename rclcpp _action::Client< ActionType >::CancelResponse::SharedPtr > >
FeedbackCallback
optional< std::chrono ::milliseconds >
shared_ptr< ActionClient >
optional< std::string >
long
vector< geometry_msgs ::msg::Pose >
vector< std::string >
#currentWaypoint_
#waypoints_
#waypointsNames_
#waypointsEventDispatcher
#onEntryThread_
#onExitThread_
#onFailure_
#onFinished_
#onSuccess_
#actionClient_
#onGoalAccepted_
#onGoalRejected_
#onNavigationFeedback_
#postNavigationAbortedEvent
#postNavigationCancelledEvent
#postNavigationSuccessEvent
#onActionFeedback_
#onNavigationAborted
#onNavigationCancelled
#onNavigationSucceeded
#postNavigationFeedbackEvent
#postAbortedEvent
#postCancelledEvent
#postSuccessEvent
#onActionAborted_
#onActionCancelled
#onActionSucceeded_
#postFeedbackEvent
#lastRequest_
#lastCancelResponse_
#feedbackCallback_
#serverTimeout_
#client_
#actionServerName
#controllerName_
#goalCheckerName_
cl_nav2z::CpWaypointNavigatorBase
+ CpWaypointNavigatorBase()
+ ~CpWaypointNavigatorBase()
+ onInitialize()
+ onStateOrthogonalAllocation()
+ loadWayPointsFromFile()
+ loadWayPointsFromFile2()
+ setWaypoints()
+ setWaypoints()
+ getWaypoints()
+ getWaypointNames()
+ getNamedPose()
+ getPose()
+ getCurrentPose()
+ getCurrentWaypointIndex()
+ getCurrentWaypointName()
+ rewind()
+ forward()
+ seekName()
+ loadWaypointsFromYamlParameter()
+ notifyGoalReached()
# insertWaypoint()
# removeWaypoint()
smacc2::SmaccAsyncClient Behavior
+ onStateOrthogonalAllocation()
+ ~SmaccAsyncClientBehavior()
+ onSuccess()
+ onFinished()
+ onFailure()
+ requestForceFinish()
+ executeOnEntry()
+ executeOnExit()
+ waitOnEntryThread()
+ onSuccess()
+ onFinished()
+ onFailure()
# postSuccessEvent()
# postFailureEvent()
# dispose()
# isShutdownRequested()
- waitFutureIfNotFinished()
cl_nav2z::components ::CpNav2ActionInterface
+ CpNav2ActionInterface()
+ ~CpNav2ActionInterface()
+ sendNavigationGoal()
+ sendGoal()
+ cancelNavigation()
+ isNavigationServerReady()
+ waitForNavigationServer()
+ onComponentInitialization()
+ onInitialize()
+ getActionClient()
+ onNavigationSucceeded()
+ onNavigationAborted()
+ onNavigationCancelled()
+ onNavigationFeedback()
- onNavigationSuccessCallback()
- onNavigationAbortedCallback()
- onNavigationCancelledCallback()
- onNavigationFeedbackCallback()
smacc2::client_core _components::CpActionClient < ActionType >
+ CpActionClient()
+ CpActionClient()
+ ~CpActionClient()
+ sendGoal()
+ cancelGoal()
+ isServerReady()
+ waitForServer()
+ onInitialize()
+ onComponentInitialization()
+ onSucceeded()
+ onAborted()
+ onCancelled()
+ onFeedback()
+ getActionClient()
- onFeedback()
- onResult()
- postResultEvent()
cl_nav2z::NavigateNextWaypoint Options
connection
ResultCode
milliseconds
bool
#client_
#navActionInterface_
#actionResult_
#goalResponseTimeout_
#resultConnectionsInitialized_
#actionClient_
#waypointname_
#options_
abortedNav2ZClientConnection
cancelledNav2ZClientConnection
succeededNav2ZClientConnection
cl_nav2z::CpWaypointNavigator
+ CpWaypointNavigator()
+ onInitialize()
+ onStateOrthogonalAllocation()
+ sendNextGoal()
+ stopWaitingResult()
- onNavigationResult()
- onGoalReached()
- onGoalCancelled()
- onGoalAborted()
smacc2::client_behavior _bases::CbActionClientBehavior Base< TAction >
+ onStateOrthogonalAllocation()
+ ~CbActionClientBehaviorBase()
+ onExit()
# sendGoal()
# cancelGoal()
# onActionSuccess()
# onActionAbort()
# onActionFeedback()
# setActionClientName()
- dispatchActionSuccess()
- dispatchActionAbort()
- dispatchActionFeedback()
#navigationResult_
#navi2ActionInterface_
< nav2_msgs::action ::NavigateToPose >
smacc2::client_behavior _bases::CbActionClientBehavior Base< nav2_msgs::action::NavigateToPose >
+ onStateOrthogonalAllocation()
+ ~CbActionClientBehaviorBase()
+ onExit()
# sendGoal()
# cancelGoal()
# onActionSuccess()
# onActionAbort()
# onActionFeedback()
# setActionClientName()
- dispatchActionSuccess()
- dispatchActionAbort()
- dispatchActionFeedback()
#waypointsNavigator_
cl_nav2z::CbNav2ZClientBehaviorBase
+ ~CbNav2ZClientBehaviorBase()
+ onStateOrthogonalAllocation()
# onNavigationActionSuccess()
# onNavigationActionAbort()
# onActionSuccess()
# onActionAbort()
cl_nav2z::CbNavigateNext Waypoint
+ CbNavigateNextWaypoint()
+ ~CbNavigateNextWaypoint()
+ onEntry()
+ onExit()
cl_nav2z::CbNavigateNamed Waypoint
+ CbNavigateNamedWaypoint()
+ ~CbNavigateNamedWaypoint()
+ onEntry()
+ onExit()